Off the Prior: Mapping Where a Synthetic-Prior Time-Series Foundation Model Fails, and Why

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Abstract

RESULTS-PENDING: the abstract is written last, from the result files.

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1. Introduction

Timestral is a family of open time-series foundation models (5M, 28M, 123M and 209M parameters) released by Mistral on October 15, 2026. It forecasts up to 1,024 steps per forward pass from up to 32,768 steps of history, outputs 21 quantiles per step, accepts past-only and known-future covariates, and was pretrained mostly on synthetic data: simple trend, seasonality and noise processes combined into multivariate series. Mistral reports first place among pretrained and zero-shot models on GIFT-Eval, TIME and fev-bench.

A model trained on a synthetic distribution learns a prior over what series look like. Benchmarks of real data measure how well that prior matches the series people happen to collect. They do not say where the prior ends. This paper asks that question directly: we build parameterized generators whose difficulty can be turned up smoothly, forecast them with all four Timestral sizes, and measure the gap to forecasters that know the true generating process. Then we look inside the model to explain the gaps, package the hard cases as benchmarks, and test whether cheap fine-tuning closes them.

Contributions:

1. **A failure map** of Timestral across four signal families (chaos and dynamics, regime switches, heavy tails and fractal processes, and a learned mixture of primitives), along every difficulty knob, at four model sizes, four context lengths and horizons from 16 steps to rollouts past one 1,024-step block, measured against a true-dynamics oracle and classical baselines.
2. **A covariate test** of Timestral's one-way covariate attention on coupled systems.
3. **A comparison of search methods** (CMA-ES, MAP-Elites quality-diversity, and gradient search through the model) for finding failures automatically.
4. **A mechanism analysis** with attention measurements on the forecast tokens and linear readouts of internal states.
5. **Two benchmarks** (a tagged diagnostic set and a leaderboard set) with generator code, seeds and a one-command evaluation.
6. **A fine-tuning study** comparing full fine-tuning, LoRA and data mixing under leave-one-family-out and held-out-knob protocols, with a regression check on normal series.

2. Setup

2.1 The model

Timestral is a causal patch transformer. A series is cut into patches of 32 steps; each patch is normalized with the mean and standard deviation of everything observed up to the end of that patch, squashed with `asinh`, and embedded. The sequence holds the context patches followed by learned forecast tokens ("Decoupled Horizon Forecasting"): each forecast token reads the whole context and the earlier forecast tokens, and a head maps it to 21 non-crossing quantiles for each of its 32 steps. Layers alternate attention over time with attention across the variates of a series. Covariate rows are read by the main-series rows but never read them. One forward pass covers up to 32 forecast patches (1,024 steps); longer horizons are produced block by block, feeding each block's median back as context. The released forecaster also averages the forecast of the series with the reflected forecast of its negation ("symmetric averaging") and clips at zero when the context is non-negative. We use the released code and weights with these defaults throughout.

Because a forecast block is causal across its forecast tokens, a forecast's first `h` steps do not depend on the requested horizon beyond `h`. We checked that slicing a 1,024-step forecast reproduces separate shorter forecasts to float precision (`tests/test_harness.py`), and evaluate all horizons up to the longest from one call.

2.2 References: what counts as a failure

A forecast error is only a failure relative to something that could have done better. We use two references:

- **True-dynamics oracle.** Every generator is a simulator with an explicit state. To build a test case we simulate a warmup and the context, copy the final state 501 times, and advance every copy over the horizon with fresh noise. Copy 0 is the realized future; the other 500 form the oracle's predictive distribution, from which we take the same 21 quantiles Timestral outputs. The oracle knows the equations, the parameters and the latent state at the forecast origin, so it is the best achievable forecast given that state (up to Monte Carlo error) and slightly stronger than anything that sees only the noisy observations. Deterministic systems get small dynamic noise and observation noise so the oracle's spread grows with the horizon instead of being zero.
- **Classical models.** Seasonal naive, ETS (additive error, damped trend, additive season when the period allows) and ARIMA (small order grid chosen by AIC), each turned into the same 21 quantiles from its Gaussian forecast distribution, fitted on the last 1,024 steps.

For every series we report the ratio of Timestral's error to the oracle's and to the best classical model on that series, and the gap (difference) to the oracle.

2.3 Metrics

- **MASE**: mean absolute error of the median, divided by the in-sample mean absolute error of the seasonal naive forecast on the context.
- **sCRPS**: the continuous ranked probability score approximated by the average pinball loss over the 21 quantile levels (times 2), divided by the same scale as MASE.
- **Coverage and calibration**: the fraction of realized values inside the 80% and 98% intervals, and the mean absolute difference between each quantile level and the fraction of realized values below that quantile.
- **WQL** (GIFT-Eval's weighted quantile loss on levels 0.1 to 0.9), reported for the leaderboard only, since it divides by the series' magnitude and breaks near zero.

Ratios are aggregated over seeds by geometric mean, with 95% bootstrap intervals.

3. Signal families

Each family is a set of simulators with a difficulty knob (full definitions in `otp/generators/`):

- **Chaos and dynamics**: Lorenz-63 (knob ρ through the onset of chaos; also the sampling interval), Rössler (knob c through period doubling), Mackey-Glass (knob: delay τ), the logistic map (knob r through the onset of chaos at about 3.57), and Kuramoto coupled oscillators (knob: coupling K through the synchronization threshold).
- **Regime switches**: jumps in mean, variance or seasonal frequency, placed inside the context at a chosen distance before the forecast origin or occurring in the horizon at a known hazard rate; and a Markov-switching AR process (knob: switching probability).
- **Heavy tails and fractal**: AR(1) and random walks with α-stable innovations (knob α), fractional Brownian motion (knob: Hurst exponent), a self-exciting (Hawkes) count process (knob: branching ratio), and a Markov-switching multifractal cascade (knob: number of multipliers).
- **Learned mixture of primitives**: a genome of four slots, each choosing one component (trend, season, AR noise, or any generator above), a weight, an additive or multiplicative role and three parameters, plus a global noise level (`otp/generators/mixture.py`). Search methods evolve genomes to maximize Timestral's error relative to the oracle while requiring the oracle to beat seasonal naive by a margin, so the series stays forecastable.
- **Controls**: in-prior series (random trend + season + AR noise, the kind of data Timestral was pretrained on).

4. Failure map

All ratios below are the geometric mean over seeds of Timestral's MASE divided by the oracle's MASE on the same series, context and horizon (1.0 = as good as the true dynamics). "vs best classical" divides by the best of seasonal naive, ETS and ARIMA on each series instead. Unless stated otherwise: context 2,048 steps, horizon 128 steps, 8 seeds per setting.

4.1 Control: series like the training data

On random trend + season + AR-noise series (the kind Timestral was pretrained on), every size stays within 0.99–1.15 of the oracle at horizons up to 128 steps. The gap grows at long horizons:

size	H=16	H=128	H=1024	H=2048
5M	1.02	1.06	1.42	2.15
28M	0.99	1.03	1.30	2.25
123M	0.99	1.01	1.12	1.36
209M	1.00	1.02	1.16	1.57

Part of that long-horizon gap is not a prior failure: the oracle knows the exact trend slope and seasonal shape, while any forecaster must estimate them from the context, and estimation error in a slope grows linearly with the horizon. Family results are read against this control.

4.2 Chaos and dynamics

series	5M	28M	123M	209M	209M vs best classical
lorenz63/rho20	1.07	1.04	1.02	1.00	0.93
lorenz63/rho28	1.31	1.28	1.27	1.24	0.95
lorenz63/rho28_oe2	29.89	30.89	27.49	27.91	0.75
rossler/c3	1.53	1.26	1.15	1.17	0.22
rossler/c5.7	3.31	2.61	2.42	2.04	0.25
rossler/c9	7.73	6.30	5.45	4.30	0.45
mackey_glass/tau10	1.47	1.32	1.22	1.16	0.11
mackey_glass/tau17	9.57	6.31	6.80	6.67	0.35
mackey_glass/tau30	21.87	21.38	12.88	13.68	0.48

series	5M	28M	123M	209M	209M vs best classical
logistic_map/r3.5	1.36	1.32	1.17	1.17	0.05
logistic_map/r3.9	1.05	1.04	1.04	1.03	0.99
kuramoto/K1	12.74	10.58	9.89	9.45	0.36
kuramoto/K8	5.18	2.28	1.86	1.43	0.11

Timestral beats the best classical model on most of these systems (the 209M model's median ratio to the best classical model across the chaos configs is 0.36) and is still far from the oracle on smooth, nearly deterministic oscillators: Mackey-Glass with long delays (21.87 at 5M, 13.68 at 209M for $\tau = 30$), weakly coupled Kuramoto oscillators (12.74 and 9.45 at K = 1), and the Rössler system (7.73 and 4.30 at c = 9). Size matters for some of these and not others: strongly coupled Kuramoto goes from 5.18 at 5M to 1.43 at 209M, while the densely sampled Lorenz system stays at 29.89 (5M) and 27.91 (209M). The chaotic logistic map (r = 3.9), whose predictability is gone after a few steps, stays near 1 at every size.

The gap lives inside the predictability horizon. A Lyapunov time is how long it takes a small error in a chaotic system to grow by a factor of e (for Lorenz at $\rho = 28$ the exponent is 0.907 per time unit). Putting four sampling intervals of the same Lorenz system on that common axis shows that the ratio is set by how many Lyapunov times the forecast reaches, not by the step count:

sampling step (time units)	horizon (steps)	horizon (Lyapunov times)	209M MASE / oracle
0.02	16	0.29	6.88
0.05	16	0.73	11.32
0.10	16	1.45	27.32
0.02	128	2.32	27.91
0.20	16	2.90	15.87
0.05	128	5.81	2.65
0.10	128	11.61	1.24
0.02	1024	18.58	1.22
0.20	128	23.22	1.18
0.02	2048	37.15	1.10
0.05	1024	46.44	1.09
0.10	1024	92.88	1.02
0.05	2048	92.88	1.06
0.10	2048	185.76	1.01
0.20	1024	185.76	1.02
0.20	2048	371.52	1.01

Inside one to three Lyapunov times the oracle can still track the trajectory and Timestral cannot; past about ten, nobody can, and the ratio returns to about 1. The noise level of the generator sets how good the oracle can be inside that window. Sweeping the noise level (dynamic and observation noise together) shows how much of each ratio that choice sets:

system	noise (% of scale)	5M	28M	123M	209M
lorenz63_rho28	0.10	1.96	1.89	1.87	1.83
lorenz63_rho28	0.30	1.65	1.60	1.57	1.55
lorenz63_rho28	1.00	1.48	1.47	1.44	1.44
lorenz63_rho28	3.00	1.24	1.21	1.19	1.18
lorenz63_rho28	10.00	1.11	1.09	1.07	1.07
mackey_glass_tau17	0.10	75.47	64.81	56.19	43.93
mackey_glass_tau17	0.30	22.32	18.29	12.69	12.68
mackey_glass_tau17	1.00	6.32	5.75	4.84	4.32
mackey_glass_tau17	3.00	2.71	2.60	2.25	2.14

system	noise (% of scale)	5M	28M	123M	209M
mackey_glass_tau17	10.00	1.76	1.81	1.48	1.46
rossler_c5.7	0.10	17.33	15.70	12.84	11.77
rossler_c5.7	0.30	10.57	9.80	7.90	6.78
rossler_c5.7	1.00	2.51	2.32	2.12	1.97

At 0.1% noise the oracle is nearly exact and Mackey-Glass sits at 43.93–75.47 times its error; at 10% the gap is 1.46–1.81. The ratio is therefore a measure of how much predictable structure Timestral leaves unused, and its size depends on how much predictable structure the generator provides. The ordering of model sizes holds at every noise level.

4.3 Heavy tails and fractal processes

Across α-stable AR processes and random walks, fractional Brownian motion, Hawkes counts and multifractal cascades, all four sizes stay between 0.94 and 1.34 of the oracle:

series	5M	28M	123M	209M	209M vs best classical
tails/stable_ar/alpha2_rho0.9	1.00	1.01	1.00	1.00	1.03
tails/stable_ar/alpha1.3_rho0.9	0.99	0.99	1.00	1.01	0.97
tails/stable_ar/alpha1.3_rho1.0	1.08	1.34	1.09	1.13	1.14
tails/fbm/H0.2	0.99	0.99	0.98	0.99	1.00
tails/fbm/H0.8	1.17	1.20	1.26	1.15	0.76
tails/hawkes/eta0.3	1.00	1.01	1.00	1.00	1.00
tails/hawkes/eta0.95	1.06	1.03	1.03	1.04	1.09
tails/msm/k8_ret	1.00	1.00	1.00	1.00	1.00
tails/msm/k8_cum	1.03	1.07	1.02	1.00	1.04

Heavy tails and long memory, which are absent from trend + season + Gaussian-noise pretraining data, do not produce failures of the size seen for deterministic dynamics.

4.4 Regime switches

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4.5 Does more history help?

All four sizes with contexts up to 32,768 steps (Colab job B, 28 settings × 8 seeds):

series	size	T=2048	T=8192	T=32768
mackey_glass/tau17	5M	9.09	6.91	6.47
mackey_glass/tau17	209M	4.91	4.54	3.84
kuramoto/K1	5M	15.26	13.86	13.74
kuramoto/K1	209M	9.47	8.10	7.62
rossler/c5.7	5M	2.83	2.59	2.49
rossler/c5.7	209M	2.23	2.10	2.06
lorenz63/rho28	5M	1.37	1.32	1.30
lorenz63/rho28	209M	1.25	1.21	1.19
regime/mean_jump/context/delta4/d64	5M	1.60	1.58	1.72
regime/mean_jump/context/delta4/d64	209M	1.08	1.10	1.13

series	size	T=2048	T=8192	T=32768
tails/fbm/H0.8	5M	1.19	1.25	1.34
tails/fbm/H0.8	209M	1.10	1.13	1.26
in_prior/p24	5M	1.05	1.12	1.35
in_prior/p24	209M	1.03	1.02	1.12

More history helps on Mackey-Glass (τ = 17: 9.09 $\rightarrow$ 6.47 for 5M and 4.91 $\rightarrow$ 3.84 for 209M from 2,048 to 32,768 steps) and slightly on Rössler and Lorenz; it does not help on the in-prior control, which gets slightly worse at 32,768 steps. 2 samples (Rössler, c = 5.7) had oracle paths that left the attractor at long horizons; their oracle ratios are excluded.

5. Covariates and one-way attention

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6. Searching for failures

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7. Why it fails

7.1 Two suspects that turned out innocent

Symmetric averaging. The released forecaster averages its forecast of the series with the mirrored forecast of the negated series. That forces a symmetric treatment of asymmetric processes (bursty counts, one-sided maps), so it was a natural suspect. Turning it off changes the sCRPS ratio to the oracle by a factor between 0.991 and 1.069 across 14 settings and both small sizes: it is not where the failures come from.

Cumulative normalization. Timestral normalizes every patch with the mean and spread of everything since the start of the context, so old history keeps weighing on the scale after a regime switch. After a variance drop 64 steps back, intervals are indeed far too wide (80% intervals cover at least 0.997 of points; sCRPS ratio 1.59–1.75). But cutting the context down to the post-switch part, which removes the old history from the statistics entirely, lowered the sCRPS ratio in only 12 of 44 cells (those with at least 256 post-switch steps) and raised it in the rest. The anchoring is not the main cause.

7.2 Template or retrieval?

For a deterministic system, the context itself is a library of past trajectories: find the moments whose recent history looks most like the present and see what happened next (the "method of analogues", a few lines of nearest-neighbour code with no training). A second context-only forecaster continues the series as a periodic pattern (the average of its last cycles at the dominant period). On the chaos family (MASE, lower is better):

series	oracle	analog	template	5M	28M	123M	209M
mackey_glass/tau17	0.29	0.93	3.85	2.79	1.84	1.98	1.95
mackey_glass/tau30	0.28	2.20	9.93	6.10	5.96	3.59	3.81
rossler/c5.7	0.48	0.81	2.02	1.59	1.25	1.16	0.98
rossler/c9	0.43	1.25	3.98	3.33	2.71	2.34	1.85
kuramoto/K1	0.20	3.54	3.96	2.61	2.17	2.03	1.94

series	oracle	analog	template	5M	28M	123M	209M
kuramoto/K8	0.06	0.08	0.87	0.32	0.14	0.12	0.09
lorenz63/rho28	1.44	1.97	2.20	1.89	1.84	1.83	1.79
logistic_map/r3.9	0.54	0.56	0.58	0.56	0.56	0.56	0.56

The method of analogues beats every Timestral size on Mackey-Glass and Rössler and loses to Timestral on weakly coupled Kuramoto and on Lorenz. Timestral's forecasts correlate more with the analogue forecast than with the periodic template, but lean toward the template more than the oracle does (`results/mechanism/analog/summary.csv`). Our reading: Timestral uses the context partly as a trajectory library and partly as a pattern to repeat, and on smooth nonlinear oscillators, which its trend + season + noise prior represents as near-periodic, the second pull wins.

7.3 Inside the model: represented but not used

For each setting we ran one plain forward pass on 300 samples and asked two questions about the oracle's expected change over the first forecast patch (and at single steps 1 and 16): how well does Timestral's own forecast predict it, and how well does a linear readout (a ridge regression, cross-validated) of the hidden state of the first forecast token predict it, at the best layer? If the readout does much better than the output, the information the oracle uses is inside the model but does not reach the forecast. The readout is a lower bound on what is linearly present: with 300 samples and 192–768 features per token, the ridge fit loses some signal, which shows on the in-prior control, where it scores 0.784–0.832 against 0.996–0.997 for the output. Gaps in the readout's favour are therefore conservative.

setting	size	MASE / oracle	output R2	best readout R2	step-1 output R2	step-1 readout R2
mean_jump_d 16	5M	3.624	0.670	0.956	0.540	0.796
mean_jump_d 16	209M	2.476	0.726	0.980	0.584	0.865
mean_jump_d 64	5M	1.536	0.919	0.962	0.803	0.841
mean_jump_d 64	209M	1.121	0.966	0.985	0.905	0.872
mean_jump_d 256	5M	1.087	0.977	0.965	0.899	0.850
mean_jump_d 256	209M	1.026	0.993	0.985	0.951	0.924
freq_jump_d 64	5M	2.269	0.894	0.960	0.084	0.684
freq_jump_d 64	209M	1.856	0.966	0.978	0.227	0.756
freq_jump_d 1024	5M	1.044	0.997	0.961	0.894	0.787
freq_jump_d 1024	209M	1.024	0.997	0.978	0.931	0.890
mackey_tau 17	5M	7.258	0.992	0.999	0.956	0.971
mackey_tau 17	209M	4.855	0.998	1.000	0.984	0.983
lorenz_rho 28	5M	1.415	0.772	0.781	0.724	0.721
lorenz_rho 28	209M	1.346	0.869	0.881	0.972	0.962
in_prior 24	5M	1.070	0.996	0.784	0.958	0.666
in_prior 24	209M	1.027	0.997	0.828	0.989	0.769

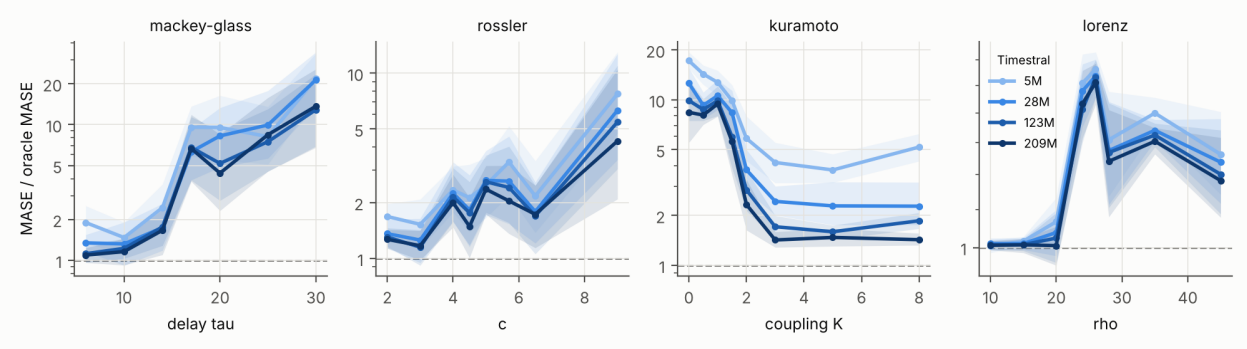


Figure 1: MASE ratio to the oracle along each chaos knob (context 2,048, horizon 128; shaded: 95% bootstrap interval over 8 seeds; dashed: as good as the oracle).

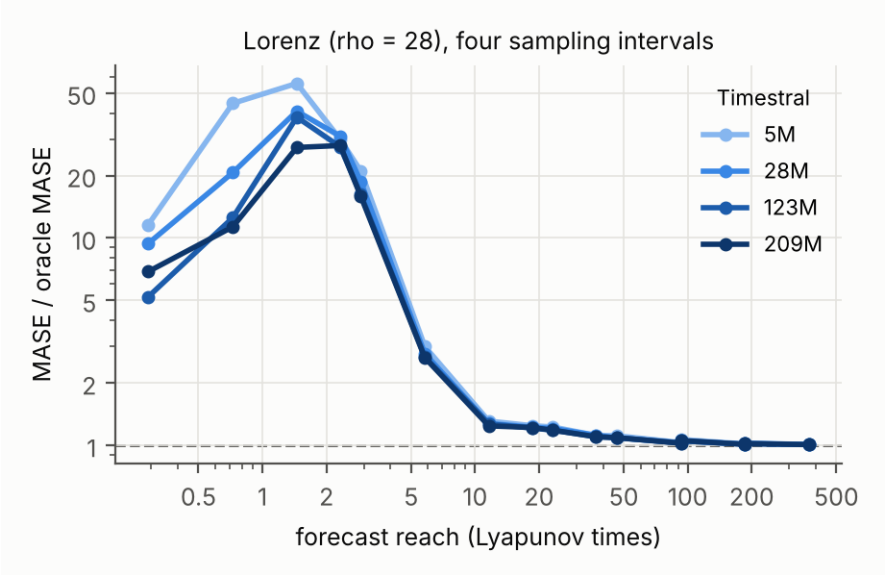


Figure 2: Lorenz at four sampling intervals on one axis: forecast reach in Lyapunov times.

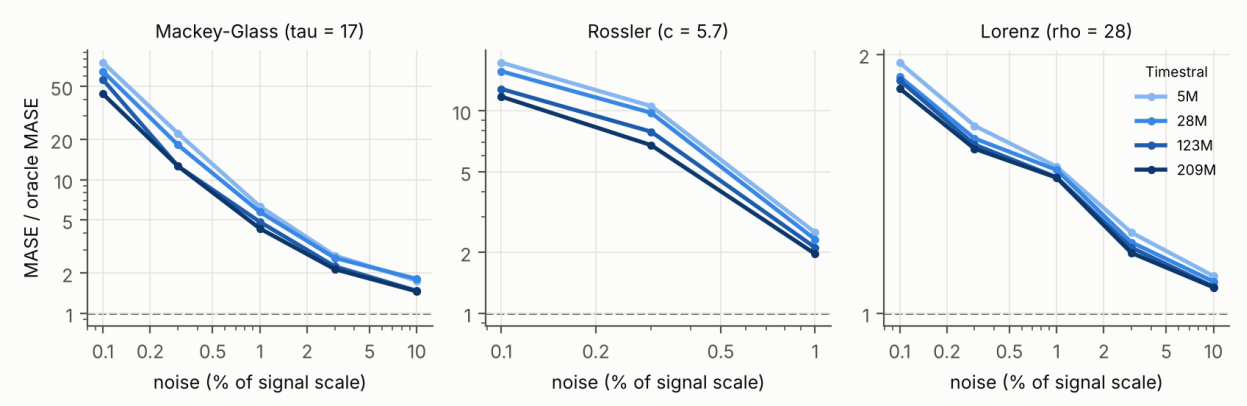


Figure 3: Ratio to the oracle against the noise level of the generator (16 seeds, horizon 128).

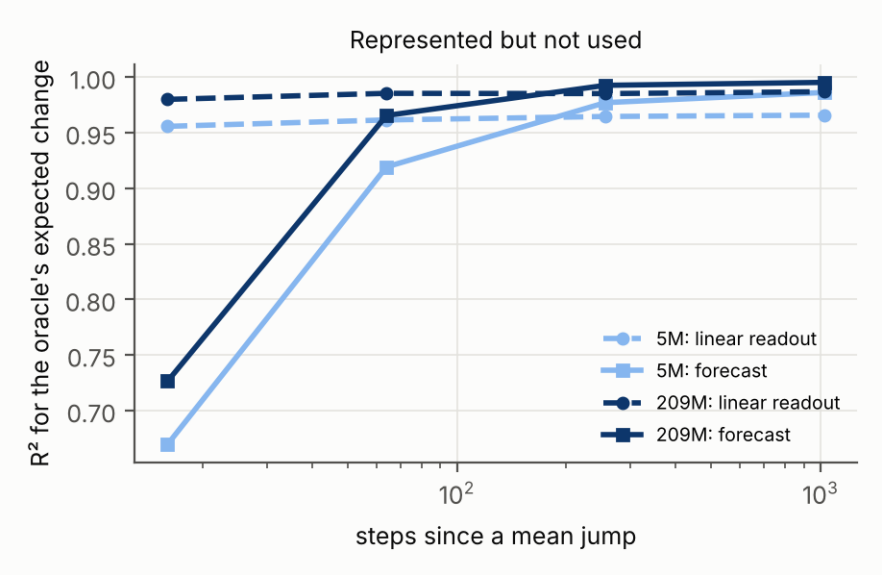


Figure 4: How well the oracle's expected change can be predicted from Timestral's forecast (solid) and from a linear readout of its first forecast token (dashed), against the time since a mean jump.

Recent regime switches: represented, attended to, not used. Sixteen steps after a mean jump, the readout reaches R^2 0.956–0.980 at every size while the forecast itself reaches 0.570–0.726, and the forecast error is 2.02–3.62 times the oracle's. The forecast tokens are not looking elsewhere: they put 15.70–17.69 times their proportional share of attention on the post-switch patches. Sixty-four steps after a frequency jump, one step ahead, the readout reaches 0.684–0.775 against 0.084–0.227 for the output. By 256 steps after a mean jump the two agree (0.977–0.993 vs. 0.965–0.985). The model sees and encodes a recent change and then forecasts as if it may not last. Our interpretation, which we cannot check without the pretraining data: if lasting breaks are rare in the synthetic trend + season + noise mix, a very recent departure from the pattern is better explained as noise under that prior than as a new regime.

Deterministic oscillators: the gap builds later. On Mackey-Glass ($\tau = 17$) output and readout agree for the first patch (0.992–0.998 vs. 0.999–1.000): the near-term forecast uses what is there, and the error accumulates over the following cycles. There the forecast correlates with a periodic continuation of the context more than the oracle's forecast does (0.87 at 5M, 0.81 at 209M, against 0.751 for the oracle), and the excess shrinks with size together with the error.

8. Benchmarks

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9. Fixing it

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10. Limitations

- **The oracle knows the latent state.** It is the best forecast given the true state, which is stronger than the best forecast given only the observed history. Ratios to it therefore bound the possible improvement from above; part of every gap is unavoidable for any forecaster that sees only the data.

- **Noise sets the size of chaos gaps.** For nearly deterministic systems the oracle's error, and so every ratio to it, depends on the noise level we chose (1% by default); Section 4.2 reports the dependence. Orderings (between settings and between model sizes) are more robust than the magnitudes.
- **Seeds.** Most settings use 8 seeds (16 for the noise sweep, 300 samples per setting for the mechanism runs). Intervals in the result files are bootstrap intervals over seeds; with 8 seeds they are wide for heavy-tailed settings.
- **Diverged simulations.** A few Rössler simulations left the attractor under noise; their oracle could not be scored and they are excluded (counts are given where they occur).
- **Univariate main series.** Except for the covariate study, every test series is a single variate.
- **No access to the pretraining data.** Statements about what the synthetic prior contains are inferences from behaviour, labelled as such.
- **Compute.** Small-model sweeps ran on a 2-core CPU VM; the large-model sweeps, the long-context runs and the mechanism runs at all sizes ran on Colab A100 GPUs; one mechanism test ran on a laptop RTX 3060 GPU. GPU and CPU forecasts of the same series agree to a few parts in 100,000 in MASE (`results/mechanism/analog/parity_vm_cpu_vs`

Reproducibility

All code, generator seeds and result files are in the repository. Every number above points to the result file it comes from; every result file has a `provenance.json` naming the script, the commit, the host and the wall time.